$$C_2H_4 + Cl_2 \longrightarrow \underset{\text{ethylene dichloride}}{C_2H_4Cl_2}$$

$$CH_4 + Cl_2 \longrightarrow CH_3Cl + HCl$$

$$C_6H_6 + Cl_2 \xrightarrow{FeCl_3} C_6H_5Cl + HCl$$

Uses of chlorine:

It is used in

1. Purification of drinking water
2. Bleaching of cotton textiles, paper and rayon
3. Extraction of gold and platinum

3.3.2 Hydrochloric acid:

Laboratory preparation:

It is prepared by the action of sodium chloride and concentrated sulphuric acid.

$$NaCl + H_2SO_4 \longrightarrow NaHSO_4 + HCl$$
$$NaHSO_4 + NaCl \longrightarrow Na_2SO_4 + HCl$$

Dry hydrochloric acid is obtained by passing the gas through conc. sulphuric acid

Properties:

Hydrogen chloride is a colourless, pungent smelling gas, easily liquefied to a colourless liquid (boiling point 189K) and frozen into a white crystalline solid (melting point 159K). It is extremely soluble in water.

$$HCl\,(g) + H_2O\,(l) \longrightarrow H_3O^+ + Cl^-$$

Chemical properties:

Like all acids it liberates hydrogen gas from metals and carbon dioxide from carbonate and bicarbonate salts.

$$Zn + 2HCl \longrightarrow ZnCl_2 + H_2$$
$$Mg + 2HCl \longrightarrow MgCl_2 + H_2$$
$$Na_2CO_3 + 2HCl \longrightarrow 2NaCl + CO_2 + H_2O$$
$$CaCO_3 + 2HCl \longrightarrow CaCl_2 + CO_2 + H_2O$$
$$NaHCO_3 + 2HCl \longrightarrow 2NaCl + CO_2 + H_2O$$

It liberates sulphur dioxide from sodium sulphite

$$Na_2SO_3 + 2HCl \longrightarrow 2NaCl + H_2O + SO_2$$

When three parts of concentrated hydrochloric acid and one part of concentrated nitric acid are mixed, Aquaregia (Royal water) is obtained. This is used for dissolving gold, platinum etc...

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